

Specialty/option: SWE-CSN-DBM

Paper: Digital Electronics

Duration: 4 hours

Credits: 7

INSTRUCTIONS:

- **Answer all questions.**
- Answer all questions. You are authorized to use only non-programmable calculator.

SECTION A: NUMBER SYSTEM AND CODES (20 MARKS)

MCQs, for each question, four answers are proposed. Write the letter for the correct answer against the question number. Each question carries 1 mark.

1. A pulse is applied to each input of an exclusive-OR gate. One pulse goes HIGH at $t = 0$ and goes back LOW at $t = 1$ ms. The other pulse goes HIGH at $t = 0.8$ ms and goes back LOW at $t = 3$ ms. The output pulse can be described as follows:
 - a) It goes HIGH at $t = 0$ and back LOW at $t = 3$ ms.
 - b) It goes HIGH at $t = 0$ and back LOW at $t = 0.8$ ms.
 - c) It goes HIGH at $t = 1$ ms and back LOW at $t = 3$ ms.
 - d) both answers (b) and (c)
2. The logic network of the Boolean function $X = (A + B)(C + D)$ is represented by
 - a) two ORs ANDed together
 - b) two ANDs ORed together
 - c) A 4-input AND gate
 - d) A 4-input OR gate
3. An example of a canonical SOP form expression is
 - a) $\bar{A}B + A\bar{B}C + AB\bar{D}$
 - b) $A\bar{B}C + \bar{A}CD$
 - c) $A\bar{B}C + ABC\bar{D}$
 - d) $A\bar{B}C\bar{D} + \bar{A}B + \bar{A}$
4. In a 4-variable Karnaugh map, a 2-variable product term is produced by
 - a) a 2-cell group of 1s
 - b) an 8-cell group of 1s
 - c) a 4-cell group of 1s
 - d) a 4-cell group of 0s
5. logic circuit with an output $X = AB + ABC$ consists of
 - a) two AND gates and one OR gate
 - b) two AND gates, one OR gate and an inverter

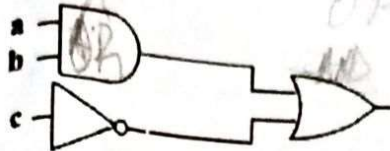
- c) two AND gates, two OR gates and two inverters
 - d) two AND gates, one OR gate and three inverters
6. A half-adder is characterized by
- a) three inputs and two outputs
 - b) two inputs and three outputs
 - c) two inputs and one output
 - d) two inputs and two outputs
7. Data distributors are basically the same as
- a) decoders
 - b) demultiplexers
 - c) multiplexers
 - d) encoders
8. If a 1-of-16 decoder with active-LOW outputs exhibits a LOW on the decimal 12 output, what are the inputs?
- a) $A_3A_2A_1A_0 = 1010$
 - b) $A_3A_2A_1A_0 = 1110$
 - c) $A_3A_2A_1A_0 = 1100$
 - d) $A_3A_2A_1A_0 = 0100$
9. If a 74HC85 magnitude comparator has $A = 1000$ and $B = 1010$, the outputs are
- a) $A > B = 0, A < B = 0, A = B = 0$
 - b) $A > B = 0, A < B = 0, A = B = 1$
 - c) $A > B = 0, A < B = 1, A = B = 0$
 - d) $A > B = 0, A < B = 1, A = B = 1$
10. J-K flip-flop with $J = 1$ and $K = 1$ has a 10 kHz clock input. The Q output is
- a) constantly HIGH
 - b) constantly LOW
 - c) a 10 kHz square wave
 - d) a 5 kHz square wave
11. A BCD-to-7 segment decoder has 0100 on its inputs. The active outputs are
- a) a, c, f, g
 - b) b, c, f, g
 - c) b, c, e, f
 - d) b, d, e, g
12. The When used with an IC, what does the term "QUAD" indicates?
- a) 2 circuits
 - b) 4 circuits
 - c) 6 circuits
 - d) 8 circuits

13. The BCD number for decimal 16 is

- a) 00010110
- b) 00010000
- c) 00010010
- d) 11100000

14. The circuit of the given figure realizes the function

- a) $Y = (a + b) \bar{c}$
- b) $Y = ab + \bar{c}$
- c) $Y = (\bar{a} + \bar{b}) c$
- d) $Y = \bar{a}\bar{b} + c$



15. The octal number $(651.124)_8$ is equivalent to

- a) $(1A9.2A)_{16}$
- b) $(1B0.10)_{16}$
- c) $(1A8.A3)_{16}$
- d) $(1B0.B0)_{16}$

16. Convert binary to octal: $(110110001010)_2 = ?$

- a) $(5512)_8$
- b) $(6612)_8$
- c) $(4532)_8$
- d) $(6745)_8$

17. What is the binary equivalent of the decimal number 368?

- a) 101110000
- b) 110110000
- c) 111010000
- d) 111100000

18. Which of the following logical operations is represented by the + sign in Boolean algebra?

- a) inversion
- b) AND
- c) OR
- d) Complementation

$$1100 = (1 \times 2^3) + (1 \times 2^2) + (0 \times 2^1) + (0 \times 2^0)$$

$$= 8 + 4 + 0 + 0$$

19. The decimal equivalent of $(1100)_2$ is

- a) 12
- b) 16
- c) 18
- d) 20

20. Any number with an exponent of zero is equal to:

- a) .zero
- b) One
- c) that number
- d) ten

SECTION B: COMPUTER FUNDAMENTAL (80MARKS)

I- Hardware (20 Marks)

- 1) Give 2 examples of computer input devices and 2 examples of output devices. For each state the use. (2 marks)
- 2) What is the function of CPU of a computer (2 marks)
- 3) Explain serial and parallel transmission of data (2 marks)
- 4) What is the meaning of SATA and IDE (2 marks)
- 5) In term of speed, how is SATA different from IDE? (3 marks)
- 6) What is the instruction set? (2 marks)
- 7) Sketch Harvard and Von-Neumann computer architecture and briefly explain the difference that exists between them. (5 marks)
- 8) What is cache memory? Why is it used in computer? (2 marks)

II- Network And Mobile Devices (40 Marks)

You have been provided with Two (02) computers and one computer that will act as the central computer to network and ensure that they should communication and share resources from a central computer

Work Required

- 1) What is a computer network? (2 marks)
- 2) What is the most appropriate OSs you think can be installed in both computers and the central computer so as to ensure that proper file sharing and resource access should be ensured. (4 marks)
- 3) List all the various networking tools you will need in course of this activity to ensure that these two computers communicate with the central machine. (5 marks)
- 4) Which other name can be given to the two computers connected to the central machine? (1 mark)
- 5) LIST all the necessary configurations you think should be put in place and the various ping commands to ensure that the computers are on a network and can communicate and share resources (8 marks)
- 6) List 4 mobile devices. (4 marks)
- 7) List 03 protocols used in mobile communication. (3 marks)
- 8) List two services available in 4G that 2G can not offer. (2 marks)
- 9) List two things you can do with a phone using its WIFI. (2 marks)
- 10) An organization is granted the block 130.56.0.0/16. The administrator wants to create 1024 subnets.
 - a) Find the subnet mask. (2 marks)
 - b) Find the number of addresses in each subnet. (2 marks)
 - c) Find the first and last addresses in subnet 1. (2 marks)
 - d) Find the first and last addresses in subnet 1024. (3 marks)

III- Microsoft Word, Excel And PowerPoint (20 Marks)

1) From the figure below:

	A	B	C	D	E	F	G	H
1								
2								
3								
4								
5								
6								
7								
8								

	LIT521	LIT529	LIT503	LIT527
Code:				
CV:	2	3	2	3

	LIT521	LIT529	LIT503	LIT527	Average	Rank	Decision
Jennifer	40	25	56	58			
Camila	45	47	58	58			

Student marks management

a) Give the formula to fill cells F7, G7 and H7 taking into account that a student is declared, "Failed" if $\text{Average} < 50$, and "Pass" if $\text{Average} \geq 50$.

(6 marks)

b) List two differences between relative references and absolute references.

(4 marks)

2) Give the suitable Office software to perform the following actions:

a) Write a letter

b) Write a financial report

c) Design a calendar

d) Prepare a presentation for a seminar

(1×4 = 4mks)

3) What is the importance of applying breaks in a document? Give the procedures.

(3 marks)

4) What is the file extension for a word, excel and PowerPoint document..

(3 marks)